

### 9.12.9. Active Tariff Components for a Timestamp

The currently active tariff components are indicated by the value of the [CurrentTariffComponents](#) attribute. Additionally, the upcoming tariff components are indicated by the value of the [NextTariffComponents](#) attribute; the value of the [CurrentTariffComponents](#) attribute will next update at the timestamp indicated by the [NextDayEntryDate](#) attribute.

Finding the active tariff components for an arbitrary timestamp requires examining the contents of the [DayEntries](#), [DayPatterns](#), [CalendarPeriods](#), [IndividualDays](#), [TariffComponents](#), and [TariffPeriods](#) attributes, and applying the following steps:

1. Determine the minutes since elapsed since midnight in the meter's timezone for the timestamp
2. For each [DayStruct](#) in the [IndividualDays](#) attribute:
  - a. If the [Date](#) field matches the timestamp:
    - i. Find the [DayEntryStruct](#) referenced in [DayStructDayEntryIDs](#) with the largest [StartTime](#) that is less than or equal to the minutes since midnight and either:
      - A. Has no [Duration](#), or
      - B. [StartTime](#) + [Duration](#) is greater than or equal to the minutes since midnight
3. If no matching [DayEntryStruct](#) was found:
  - a. Determine the day of the week for the timestamp
  - b. Find the [CalendarPeriodStruct](#) in the [CalendarPeriods](#) attribute with the largest [StartDate](#) that is less than or equal to the timestamp
  - c. If a matching [CalendarPeriodStruct](#) was found:
    - i. For each [DayPatternStruct](#) in the [DayPatterns](#) attribute referenced in [DayPatternIDs](#):
      - A. If the bit matching the day of the week is set in [DaysOfWeek](#):
        - I. Find the [DayEntryStruct](#) referenced in [DayStructDayEntryIDs](#) with the largest [StartTime](#) that is less than or equal to the minutes since midnight
4. If a matching [DayEntryStruct](#) was found:
  - a. For each [TariffPeriodStruct](#) in the [TariffPeriods](#) attribute:
    - i. If the [DayEntryIDs](#) field contains the [DayEntryID](#) of the matched [DayEntryStruct](#), then the [TariffComponentIDs](#) field contains the [TariffComponentIDs](#) of the active [TariffComponentStructs](#).
5. Otherwise, there is no available information in the tariff for determining the tariff components active at the timestamp